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Design and Development of a Device to Harvest AC Compressor Exhaust Heat for Sustainable Electrical Energy Production

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***Abstract*—The increasing demand for air conditioners raises the problem of thermal energy wastage. Domestic air conditioners waste low-grade heat of 30–70 °C, which cannot be used efficiently by classical heat-to-power methods. This paper presents an approach to the sustainable conversion of this heat energy into usable electrons, stored in an electric cell. Experimental research has been performed on two different materials, stainless-steel and copper plates, with thermal adhesive to optimize physical conduction into a Peltier module. The temperature difference between the two sides of the module creates the need for an optimal cold-side heat sink to maximize domestic on-site power generation. The developed algorithms are applied to real-time data, and the outcomes are compared to identify the better efficiency, as proved through data visualization and graphical representation.**

***Index Terms*—Sustainable-energy; Heat-harvesting; Charged-Electrons.**

I. Introduction

AC (air-conditioning) systems are increasingly used in residential and commercial buildings, and a large amount of the electricity they use is ultimately rejected as waste heat through the condenser unit. Typically, several hundred watts of thermal energy are persistently transferred to the ambient environment during normal operation, even in small domestic air conditioners. This waste heat falls within the low-temperature range (approximately 30–70 °C) and therefore remains largely unutilized due to its limited potential for traditional energy conversion methods. Recovering energy from this low-grade thermal source without disturbing the primary cooling performance of the air-conditioning system poses a challenge [1]. Peltier modules operating in power generation mode offer a potential pathway to address this issue, as they can directly convert potential/temperature differences into electrical energy.

The main conversion stage makes use of the Peltier module, functioning as a thermoelectric generator (TEG), producing a direct-current (DC) voltage. The energy output of a Peltier module is mainly dependent on the temperature difference (ΔT) maintained across its two ceramic sides. As the hot side absorbs waste heat, the system needs to dissipate heat from the other side using a heat sink to prevent thermal equilibrium, which would stop energy generation, as the TEG relies on the thermal potential difference.

However, the direct use of a Peltier module in this application presents electrical challenges due to the low output current and voltage fluctuations of the Peltier module. To address this, the paper suggests using the harvested energy to charge a solid-state battery (SSB). SSBs are well-suited for this application due to their high safety profile and energy density, making them an ideal candidate for domestic energy storage systems, although any electric cell can be used to store the generated electrons [2].

Domestic refrigerators and air conditioners are the most widely used household appliances worldwide; however, they are infrequently acknowledged as potential sources of usable thermal energy. When operated, these appliances continuously emit heat through compressor shells and condenser coils to the surrounding environment. The rejected thermal exhaust usually comes out as a steady flow of low-temperature hot air, representing an underutilized energy resource. Considering the massive global deployment of such appliances, even small amounts of recovered waste heat can add up to significant energy savings. Practical implementation poses difficulties

owing to variations in compressor size, geometry, and installation constraints, motivating the exploration of adaptable and retrofittable heat-capture mechanisms.

This paper presents an effective way to use the waste thermal energy rejected by domestic air-conditioner exhaust systems to produce electricity, providing the combined advantages of cost-free electricity production and mitigation of thermal pollution. By converting low-grade heat that would otherwise be dissipated into the ambient environment, the proposed system contributes toward minimizing unnecessary thermal emissions, which indirectly intensify the accumulation of urban heat and contribute to global warming. It presents a practical implementation of low-temperature thermal energy harvesting using thermoelectric conversion, validated through an assessment of thermal pathways and system components. The work highlights the potential for integrating waste heat recovery mechanisms into everyday household appliances, thereby offering a platform for subsequent innovations in residential-scale sustainable energy recovery technologies.

A. Background

1) Thermodynamic constraints in low-grade heat recovery: The primary challenge in recovering energy from domestic air-conditioning systems lies in the limited temperature difference (ΔT) between the heat source and the ambient environment. According to Carnot's theorem, the maximum theoretical efficiency of any heat-to-power conversion system is directly proportional to ΔT. In domestic settings, the hot-side temperature typically ranges between 30 °C and 70 °C, while the ambient air temperature remains around 25–35 °C, resulting in inherently low conversion efficiency. Consequently, prior research emphasizes that system optimization under such constraints must focus on minimizing heat losses rather than increasing the source temperature [3], [4].

2) Thermal interface resistance and material considerations: Several studies have reported that thermal interface resistance significantly affects the performance of thermoelectric energy harvesting systems operating at low temperature gradients. Microscopic air gaps between the heat source and harvesting surfaces act as thermal insulators, severely degrading heat transfer efficiency. To mitigate this issue, the use of thermal interface materials (TIMs) has been widely explored to improve surface contact and reduce interfacial thermal resistance [5], [6].

B. Existing research

Thermoelectric devices have attracted considerable attention because of their versatility in cooling, dehumidification, heat transfer, and energy conversion applications. Recent studies have integrated Peltier modules with air-conditioning systems for cooling enhancement and water recovery, demonstrating their suitability for sustainable energy-related applications [7].

Nohra et al. constructed a thermal model and examined the geometric parameters of TEG plates and duct dimensions to maximize output. However, the integration complexity and the dependence on matching HVAC and engine conditions can restrict real-world adoption for small appliances [8].

Roungbungrud et al. demonstrated a prototype in which condenser heat is utilized more effectively for electricity generation. It captures waste heat from a split AC condenser and cools the module using the condensate water that drips off. The prototype yielded very low power due to the small ΔT and limited heat transfer area [9].

Hasalli Ibrahim et al. analyzed and built a TEG model specifically for AC waste heat, comparing air-cooled and water-cooled configurations to determine which produced more electricity when tested at simulated hot-side temperatures. Even with optimized cooling, the real power remained low due to the very small temperature gradients in domestic systems [10].

II. Methodology

A. Working principle of thermoelectric power generation

Thermoelectric conversion is a technique for turning a temperature difference directly into electricity without moving parts or working fluids. This is mainly based on a phenomenon called the Seebeck effect. When a temperature difference is established across a TEG module, charge carriers migrate from the hot side to the cold side within the semiconductor material, creating an electrical potential difference. The voltage is proportional to the temperature difference (ΔT) and to the Seebeck coefficient of the materials. When an electrical load is connected, a current flows, and heat is converted into electrical power [11], [12].

The Seebeck voltage generated by the thermoelectric module is expressed as:

$$V = S\Delta T \tag{1}$$

where V is the generated voltage, S is the Seebeck coefficient, and ΔT is the temperature difference between the hot and cold sides of the module.

Electrical power generated by the thermoelectric module can be calculated using:

$$P = VI \tag{2}$$

where P is the output power, V is the output voltage, and I is the output current. Maximizing power generation requires maintaining an optimal temperature difference across the module, as a larger effective temperature difference leads to improved electrical output.

B. Experimental setup

Six Peltier modules were used in this experimental setup to convert waste heat from an air-conditioning system into electrical energy. The heat rejected by the air conditioner served as the hot-side thermal source for the thermoelectric modules. Two different heat transfer materials, copper and stainless steel, were investigated to evaluate their influence on thermoelectric performance. These materials were placed between the waste heat source and the thermoelectric modules to facilitate heat transfer.

Figure 1 shows the setup, in which the hot side of the thermoelectric modules was attached to the heat transfer plate, while the cold side was connected to a heat sink exposed to ambient air. The setup preserved a temperature gradient across the modules, thereby supporting the generation of thermoelectric power. Electrical measurements, including voltage and current, were recorded at regular time intervals. The generated power was subsequently calculated using the measured voltage and current values.

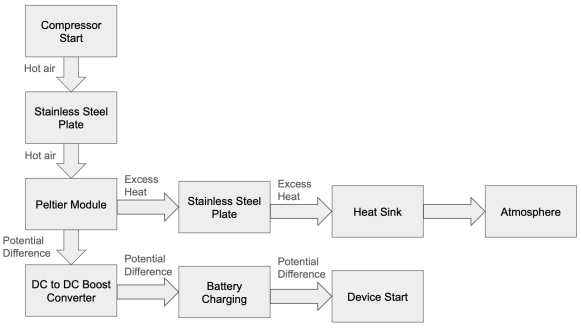
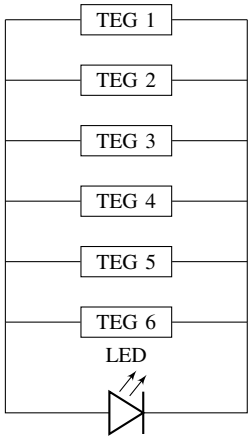


Fig. 1. AC compressor as thermoelectric heat generator

1) Parallel configuration



2) Series configuration

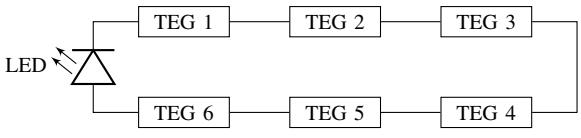


Fig. 2. Circuit diagrams of the parallel and series configurations using six Peltier modules

C. Experimental Procedure

The experiment was conducted under uniform conditions for all configurations. Initially, six Peltier modules were mounted between the heat transfer plate and the heat sink. Waste heat was allowed to flow from the air-conditioning system through the selected plate material to the hot side of the Peltier modules.

The air-conditioning system was operated continuously for each test configuration while voltage and current readings were recorded at predefined time intervals. Data were measured for two configurations, series and parallel, using the copper and stainless-steel plates. The data comprise voltage-versus-time, current-versus-time, and power-versus-time readings, which are plotted as graphs in the following section.

III. Results

The experimental results for the two plates, copper and stainless steel, are presented below in graphical form in Figures 3 to 17. The plates were tested in two configurations, series and parallel, and the graphical outputs are presented in three forms: voltage, current, and power.

A. Effect of stainless-steel plates

1) Series configuration: Figure 3 demonstrates a gradual increase in voltage with time, reaching a stable region and indicating effective thermoelectric generation under continuous heating. A steady rise in current was observed in Figure 4, signifying improved transport of charge carriers with the development of the temperature gradient. Confirming successful thermal-to-electrical energy conversion, the power output shown in Figure 5 increased with heating duration and reached its maximum value at higher operating temperatures.

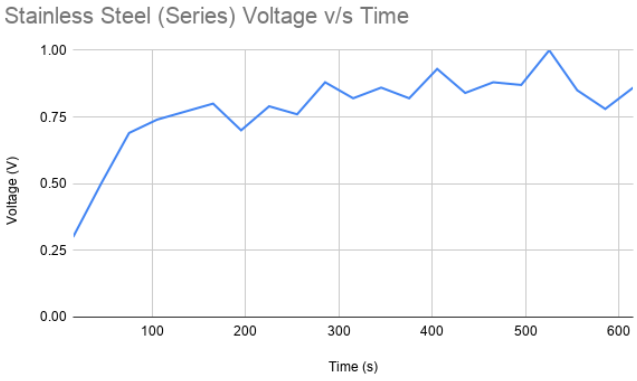


Fig. 3. Voltage vs. Time: stainless-steel plates connected in series

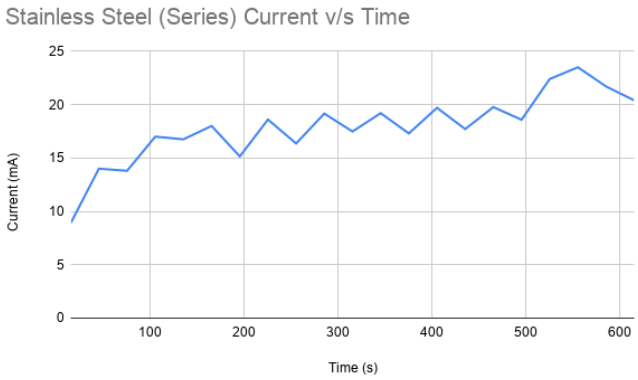


Fig. 4. Current vs. Time: stainless-steel plates connected in series

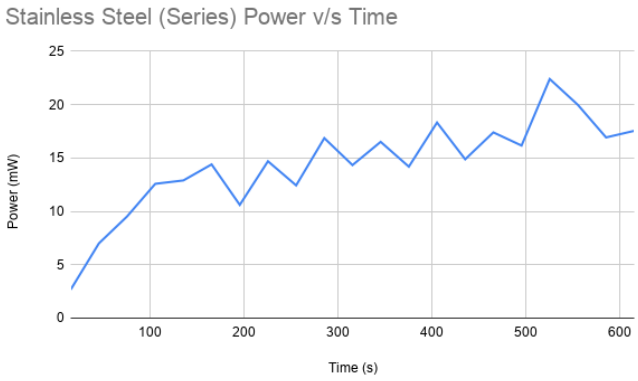


Fig. 5. Power vs. Time: stainless-steel plates connected in series

2) Parallel configuration: The voltage shown in Figure 6 remained relatively low and stable throughout the experiment due to the characteristics of a parallel connection. The current (Figure 7) initially increased and later stabilized, indicating moderate thermoelectric performance. Power generation, shown in Figure 8, was inferior to that of the series setup, reflecting decreased effectiveness in harvesting energy.

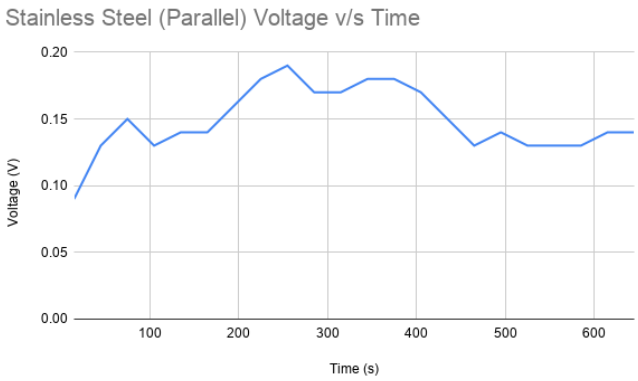


Fig. 6. Voltage vs. Time: stainless-steel plates connected in parallel

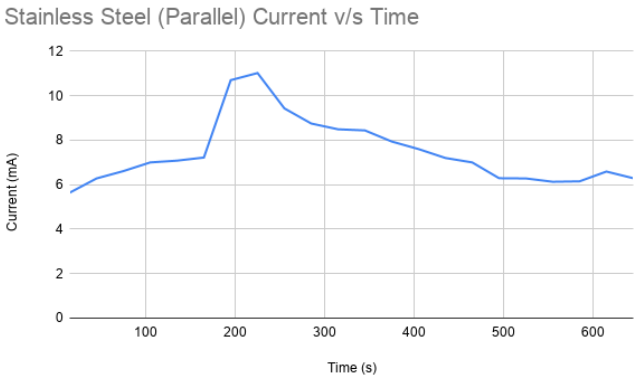


Fig. 7. Current vs. Time: stainless-steel plates connected in parallel

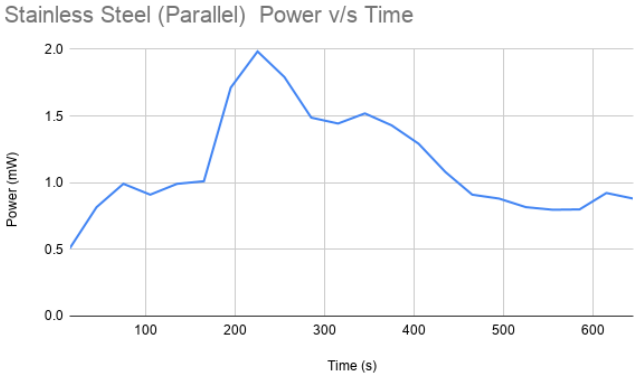


Fig. 8. Power vs. Time: stainless-steel plates connected in parallel

B. Effect of copper plates

1) Series configuration: Copper plates produced a stable voltage output during operation (shown in Figure 9). Figure 10 presents the current, which increased gradually over time and subsequently leveled off. The resulting power output, shown in Figure 11, demonstrated moderate energy conversion capability, though peaking at lower sustained levels compared to the stainless-steel configuration.

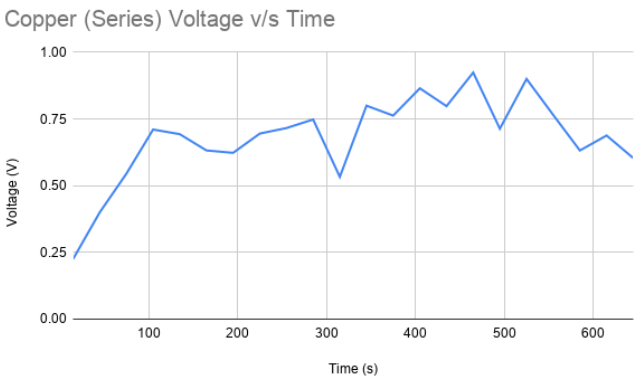


Fig. 9. Voltage vs. Time: copper plates connected in series

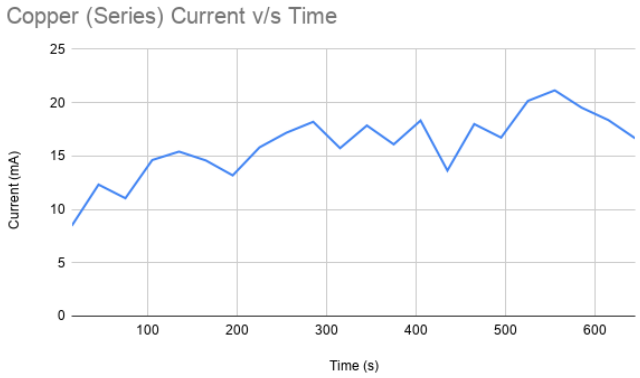


Fig. 10. Current vs. Time: copper plates connected in series

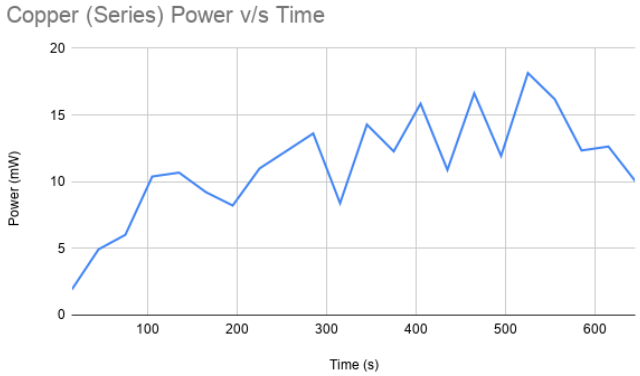


Fig. 11. Power vs. Time: copper plates connected in series

C. Copper and stainless steel in series

Figures 12, 13, and 14 present the direct comparative analysis of both materials in the series configuration. While both materials exhibited similar trends during initial operation, stainless-steel plates achieved slightly higher and more stable voltage generation across the testing duration. In the current measurements, stainless steel consistently produced a marginally higher current output than copper over the majority of the operating period. Consequently, stainless steel generated superior overall output power, demonstrating higher energy conversion efficiency within this system configuration.

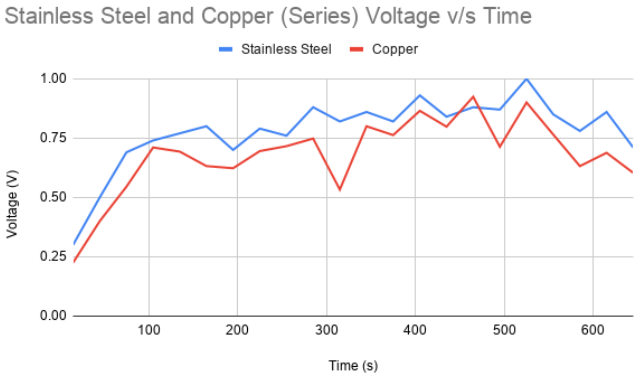


Fig. 12. Voltage vs. Time: comparison between stainless-steel and copper plates in series

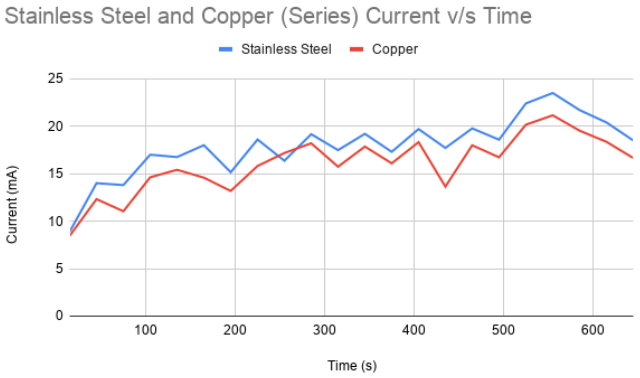


Fig. 13. Current vs. Time: comparison between stainless-steel and copper plates in series

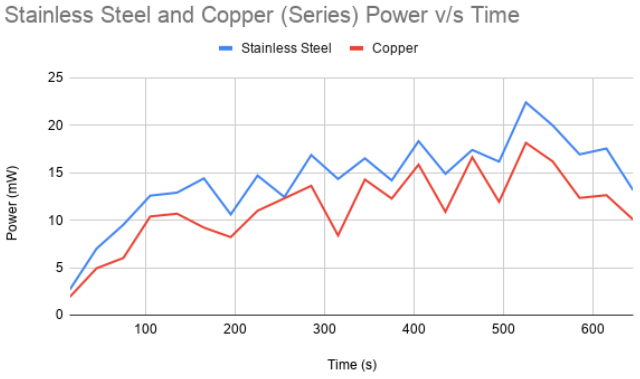


Fig. 14. Power vs. Time: comparison between stainless-steel and copper plates in series

D. Series vs parallel configuration

Figures 15, 16, and 17 show that the series configuration generated a significantly higher voltage than the parallel configuration. A comparatively greater and steadier current was recorded in the series arrangement. In power generation, the

series configuration also produced substantially greater power output, demonstrating superior energy harvesting performance.

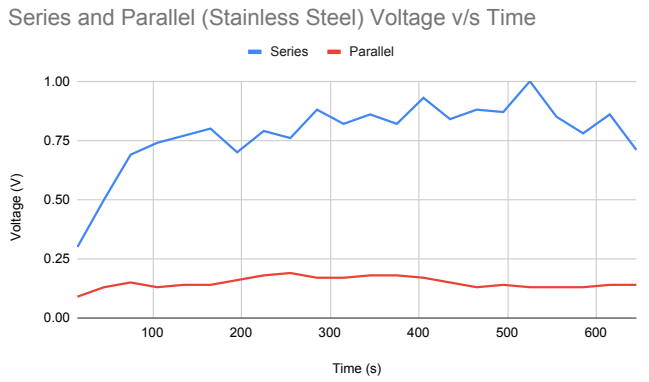


Fig. 15. Voltage vs. Time: stainless-steel plates, comparison between series and parallel

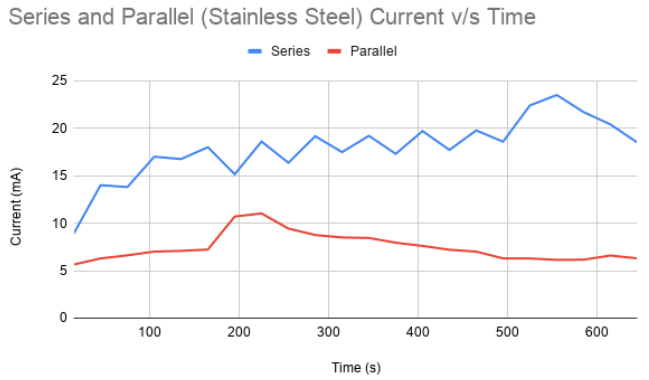


Fig. 16. Current vs. Time: stainless-steel plates, comparison between series and parallel

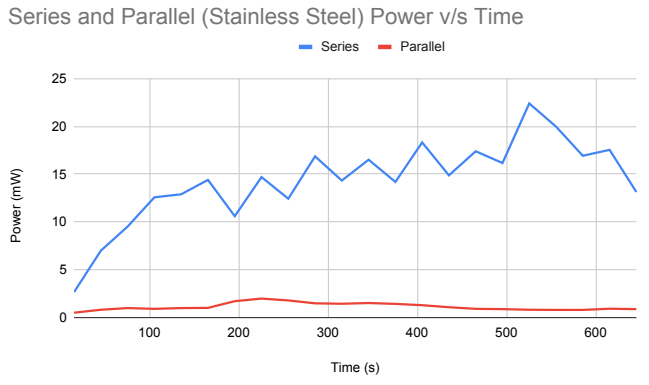


Fig. 17. Power vs. Time: stainless-steel plates, comparison between series and parallel

Based on the experimental results obtained from the voltage, current, and power measurements, the stainless-steel plates in the series configuration demonstrated the best overall performance among all tested arrangements, as summarized in Table I. Stainless steel consistently produced higher and more stable voltage and current outputs, resulting in improved power generation.

TABLE I
COMPARATIVE ANALYSIS OF INTERFACIAL PLATE MATERIALS (SERIES SETUP)

Parameter	Stainless steel	Copper	Superior Configuration
Thermal conductivity	Lower (≈15 W/m·K)	Higher (≈400 W/m·K)	Copper (Material property)
Maintained hot-side ΔT	Higher	Lower	Stainless steel
Voltage output	Higher	Lower	Stainless steel
Current output	Higher	Lower	Stainless steel
Power output	Higher	Lower	Stainless steel
Energy harvesting efficiency	Superior	Moderate	Stainless steel

The improved electrical output observed with stainless steel can be attributed to its lower thermal conductivity relative to copper, which helped maintain a higher localized temperature gradient (ΔT) across the Peltier modules. In this passive, air-cooled heat sink arrangement, copper's rapid thermal dissipation caused the cold-side heat sink to saturate thermally at a faster rate, reducing the net temperature gradient across the module faces. Conversely, stainless steel facilitated a sustained thermal differential, directly translating into greater Seebeck voltage and overall electrical power. Furthermore, series-connected modules produced markedly greater electrical performance than parallel arrangements, making series the optimal configuration for thermoelectric energy harvesting applications.

IV. CONCLUSION

This paper demonstrated the feasibility of converting waste heat from an air-conditioning system into electrical energy using thermoelectric modules. The results of the experiments conducted using six Peltier modules showed that performance metrics such as voltage, current, and power were considerably influenced by the thermal interface material and the operational arrangement. Of the two investigated materials, stainless steel exhibited superior performance compared to copper, sustaining a larger temperature gradient across the modules and resulting in greater electrical output. The findings verify that thermoelectric energy harvesting can convert waste heat into useful power, offering a practical, green, and sustainable method for low-power generation. Future work may focus on improving conversion efficiency through optimized module arrangements and advanced heat transfer techniques.

V. FUTURE SCOPE

The performance of the proposed thermoelectric energy harvesting system can be further improved by employing advanced thermoelectric materials with higher conversion efficiency and optimizing the hot-side and cold-side thermal management. The ultimate goal of this experiment is to build a small, easy-to-install module compatible with existing systems, enabling large-scale deployment. With further optimization, such systems could aid in mitigating thermal pollution while enhancing energy utilization within residential environments.

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